Course Title:	Financial Modeling and Automation
Apogee Code:	MMMCA106
Period:	Semester 1
ECTS Credits:	2
Language of Instruction:	English
Delivery Mode:	On campus
Total of Hours:	37
Hours of Lectures:	0
Hours of Tutorials:	15
Personal Work Hours / Student:	22
Instructor(s):	Moncef GAROUANI / Pierre-Paul CAVALLERA
Teaching Assistant(s):	

Presentation

Computers and digital transactions have become evermore important in the financial industry. Algorithms and models are the driving forces behind the software used for this ever-increasing automatization.
In this course, students learn the basics of programming, how to construct algorithms and work with structured data. The entire course is given on a hands-on approach where the students code together with the instructor.

Link with the Programme

The course contributes to the following program ILOs:

- tackle new and complex problems related to finance in a professional context,
- use IT tools in financial applications.

The course is a prerequisite for the course of Machine Learning and Big Data.

Disclaimer
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Learning Objectives

By the end of the course, students shall be able to:

- apply key notions of programming to Python — these notions are applicable to many other programming languages than Python,
- write, document and test simple Python programs,
- read in and manipulate structured data using dedicated libraries,
- synthesize information and present the results in a written form.

Transversal Skills

Managerial Skills:

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International: no

Ethics, Responsibility, Sustainability: no

Prerequisite

Laptop (MacOS, Windows or Linux) with Spyder installed.
Basic knowledge of computer functions.

Content, Organisation

1 - About this course
2 - Python for finance: what and why?
4 - Interacting with Python
5 - Getting started
6 - Basic control structures
7 - Variables, predefined data types and related operators
8 - Basic data structures
9 - Libraries
10 - Advanced notions

Guidelines for maintaining a positive classroom environment:

- Your questions are always welcome, though I may defer the answers.
- We will start each lesson on time; late arrivals cause interruptions in the flow of the course.

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Content, Organisation

- If you do not attend a lesson, do not forget to inform the administration and to provide appropriate justifications; being late or missing will impact negatively your continuous assessment grade i.
- Deadlines must be respected; work provided later that requested will be examined or not depending on my workload and remaining time, penalties to be expected.

Assessment

Continuous Assessment: 50%
Final exam : 50%

Type of Assessment

Continuous assessment: 50%

- Homework and in class exercises: 40% (16/20)
  Some exercises are evaluated without prior warning.
  All resources are permitted. However, such tools should be used for learning purposes, not to avoid the homework, as they will not be available during final exams.
- Participation: 10% (4/20)
  The first unexcused absence costs 2 points out of 4.
  With more than one unexcused absence, you get 0 points out of 4.
  Cumulated lateness is taken into account.

Final computer exam: 50%

Exercises on Python programming.
Training material and personal notes allowed.
Access to internet and AI assistants strictly prohibited

Bibliography

Python tutorial: https://docs.python.org/3/tutorial/index.html
Turtle: https://docs.python.org/3/library/turtle.html
Pandas: https://pandas.pydata.org/docs/user_guide/index.html
Numpy: https://numpy.org
Pyplot: https://matplotlib.org/stable/tutorials/pyplot.html#pyplot-tutorial
Spyder: https://www.spyder-ide.org

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